

Another common notation for performing computations involving Hamiltonian systems is the *Poisson bracket* which is defined for two smooth scalar-valued functions g_1 and g_2 of the phase variables $(\mathbf{q}, \mathbf{p})$ of a Hamiltonian system in $\mathbb{R}^m$ by

$$\{g_1, g_2\} = \sum_{i=1}^{N_c} \left(\frac{\partial g_1}{\partial q_i} \frac{\partial g_2}{\partial p_i} - \frac{\partial g_2}{\partial q_i} \frac{\partial g_1}{\partial p_i} \right) = \nabla g_1^T \mathbf{J} \nabla g_2,$$

where $\mathbf{J}$ is the skew symmetric symplectic structure matrix:

$$\mathbf{J} = \begin{bmatrix} 0 & \mathbf{I} \\ -\mathbf{I} & 0 \end{bmatrix},$$

and $\mathbf{I}$ represents the $N_c \times N_c$ identity matrix. The Poisson bracket has the following properties, which are easily verified from the definition (for g_1, g_2, g_3 being three arbitrary functions of the phase variables):

Bilinearity $\{g_1, \alpha g_2 + \beta g_3\} = \alpha \{g_1, g_2\} + \beta \{g_1, g_3\}$.

Skew symmetry $\{g_1, g_2\} = -\{g_2, g_1\}$ and this implies $\{g_1, g_1\} = 0$.

Jacobi identity $\{g_1, \{g_2, g_3\}\} + \{g_3, \{g_1, g_2\}\} + \{g_2, \{g_3, g_1\}\} = 0$.

In terms of the Poisson bracket, it is possible to write the differential equation corresponding to a coordinate q_i , say, as

$$\dot{q}_i = \{q_i, H\}.$$

More generally, if $F(\mathbf{q}, \mathbf{p})$ is any smooth, scalar-valued function of the phase variables, we may write

$$\dot{F} = \frac{d}{dt} F(\mathbf{q}(t), \mathbf{p}(t)) = \{F, H\}.$$

As a consequence, we have the following relation between the Lie derivative and the Poisson bracket:

$$\mathcal{L}_{\mathbf{J}\nabla H} F \equiv \{F, H\}.$$

For a Hamiltonian flow, we typically simplify notation by writing $\mathcal{L}_H$ in place of $\mathcal{L}_{\mathbf{J}\nabla H}$.

3.3 Backward Error Analysis for Hamiltonian Splitting Methods

As the relation

$$\mathcal{L}_H \phi = \{\phi, H\}$$

is linear in H , we may write, for Hamiltonians H_1, H_2 ,

$$\mathcal{L}_{H_1+H_2} = \mathcal{L}_{H_1} + \mathcal{L}_{H_2}.$$

The flow map of the system with Hamiltonian $H = H_1 + H_2$ is

$$\mathcal{F}_t = e^{t(\mathcal{L}_{H_1} + \mathcal{L}_{H_2})}.$$

On the other hand, the splitting method based on a composition of flows on H_1 and H_2 is

$$\mathcal{G}_h = e^{h\mathcal{L}_{H_1}} e^{h\mathcal{L}_{H_2}}.$$

The issue then becomes the comparison of the exponential series

$$e^{h(A+B)}$$

with

$$e^{hA} e^{hB},$$

for noncommuting operators A, B . Expanding these out using the exponential series we have

$$\begin{aligned} e^{h(A+B)} &= \text{Id} + h(A+B) + \frac{h^2}{2}(A+B)^2 + \frac{h^3}{6}(A+B)^3 + \mathcal{O}(h^4) \\ &= \text{Id} + h(A+B) + \frac{h^2}{2}(AB+BA+A^2+B^2) \\ &\quad + \frac{h^3}{6}(A^3+A^2B+AB^2+ABA+B^2A+BA^2+BAB+B^3) \\ &\quad + \mathcal{O}(h^4), \end{aligned}$$